Hallucination Fingerprints: Consistent Failure Patterns in Large Language Models

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Abstract

When a language model confidently states that the capital of Germany is Paris, something has gone wrong inside the model before that word ever appears. This paper investigates what. We ask whether hallucinations follow *consistent, detectable patterns* in the internal activations of transformer models prior to the generation of an incorrect token — and find that they do. Through experiments on a custom 806K-parameter transformer and GPT-2 (124M parameters), tested across 20,000 factual prompts in 7 knowledge categories, we identify two named phenomena: **Relation Dropout**, in which attention to the semantic relation token collapses in the final transformer block of small models; and **Last-Layer Suppression**, in which factual knowledge emerges correctly in blocks 10–11 of GPT-2 but is systematically overridden by block 12. We propose a three-type hallucination taxonomy, release **HallScan** (`pip install hallscan`), an open-source detection tool, and **HallBench**, a labeled benchmark of 20,000 annotated examples.

1 Introduction

Consider the following exchange with GPT-2:

"The capital of Germany is" → **"the"** (confidence: 89.1%)

The model is wrong. But more striking than the error is the confidence. GPT-2 does not hedge. It does not say "I think." It predicts "the" as though Berlin were never a possibility.

Yet Berlin *was* a possibility — inside the model. Our layer-by-layer analysis reveals that at block 11, the probability of "Berlin" reaches 34.7%. The model has found the answer. Then block 12 erases it, and "the" wins with 41.2% probability. The model knew. Then it forgot.

This paper is about that forgetting. Specifically, we ask: *do language models hallucinate in consistent, detectable patterns visible in their internal activations before the incorrect token is generated?*

If such patterns exist — and we show that they do — they could form the basis of a new class of hallucination detectors that intercept errors during generation rather than catching them post-hoc.

Contributions. We make the following contributions:

- We identify and name **Relation Dropout**: in small transformer models, attention to the semantic relation token (e.g., "capital") collapses in the final block immediately before a hallucination, even when the entity token (e.g., "germany") is strongly attended to.

- We identify and name **Last-Layer Suppression**: in GPT-2, factual knowledge emerges with high probability in blocks 10–11, then block 12 systematically suppresses this signal in favour of high-frequency structural tokens. This pattern holds across all 20,000 tested prompts, with an average suppression layer of exactly 12.0.
- We propose a **three-type hallucination taxonomy** (Type 1: Relation Dropout; Type 2a: Last-Layer Suppression; Type 2b: Knowledge Gap) grounded in internal activation analysis rather than output-level evaluation.
- We discover **prompt sensitivity** as a measurable phenomenon: the same factual query phrased differently produces 8.1% accuracy in one form and 0.0% in another, suggesting GPT-2 has learned surface-form patterns rather than abstract facts.
- We validate our findings at scale across **20,000 factual prompts** in 7 knowledge categories on an NVIDIA RTX 4060 GPU.
- We release **HallScan** (`pip install hallscan`), an open-source Python library for hallucination detection, and **HallBench**, a labeled benchmark of 20,000 annotated examples on HuggingFace.

2 Related Work

Hallucination in NLG. Ji et al. [2023] provide a comprehensive taxonomy of hallucination in natural language generation, distinguishing intrinsic errors (contradicting source material) from extrinsic ones (unverifiable claims). Lin et al. [2022] introduce TruthfulQA, demonstrating that larger models are not necessarily more truthful — a counterintuitive finding that motivates our investigation of internal mechanisms rather than scale.

Factuality and calibration. Kadavath et al. [2022] study model self-knowledge, finding that well-calibrated models can identify their own uncertainty. Our work complements this by showing that even when a model "knows" the answer (as evidenced by high probability in intermediate layers), output-level calibration may still fail due to Last-Layer Suppression.

Mechanistic interpretability. Elhage et al. [2021] establish a mathematical framework for transformer circuits. Olsson et al. [2022] identify induction heads as key in-context learning mechanisms. We extend this interpretability tradition to the specific problem of factual hallucination, examining which components fail and at which layer.

Knowledge in FFN layers. Geva et al. [2021] demonstrate that feed-forward layers function as key-value memories. Our finding that factual knowledge emerges in blocks 10–11 before being overridden by block 12 is consistent with this view: the FFN layers may correctly retrieve factual associations, which are then suppressed by a final output-selection mechanism.

3 Method

3.1 Models

Custom transformer (806K parameters). We train a GPT-style decoder-only transformer with 4 blocks, 4 attention heads, and $d_{\text{model}} = 128$ on a curated corpus of factual capital-city and geographic statements (e.g., "the capital of france is paris"), repeated to produce 8,350 training tokens. We train for 10 epochs using the Adam optimiser with learning rate 3×10^{-4} and observe loss converging to near zero — demonstrating overfitting to the training distribution, which allows us to study hallucination in a fully controlled environment.

GPT-2 (124M parameters). We use the standard GPT-2 model [Radford et al., 2019] pretrained on 40GB of internet text (WebText). This model serves as our production-scale validation environment, representative of the class of autoregressive language models deployed in practice.

3.2 Internal Activation Capture

For each prompt, we capture:

1. **Attention weights** in every block averaged across all heads, with particular focus on attention directed to the relation token.
2. **Hidden state projections** at each block, obtained by passing the residual stream through the model's final layer norm and unembedding matrix. This allows us to compute the implied vocabulary distribution at each intermediate layer — revealing what the model "would predict" if generation stopped at that block.
3. **Top-10 predictions** and their probabilities at the final output layer.

3.3 Hallucination Taxonomy

We classify each incorrect prediction into one of three types:

Type 1 — Relation Dropout. Average attention to the relation token in the final block falls below $\tau = 0.05$. The model fails to route attention to the semantically critical word that constrains the answer type. Characteristic of small, undertrained models.

Type 2a — Last-Layer Suppression. The correct answer appears in the top-10 predictions at the final output layer but is not the top-1. Internal layer analysis reveals the correct answer peaked in blocks 10–11 before being suppressed by block 12.

Type 2b — Knowledge Gap. The correct answer does not appear in the top-10 predictions. The relevant fact was not encoded during training.

4 Experiments

4.1 Experiment 1: Relation Dropout in Small Models

We test our 806K-parameter model on 6 capital-city prompts and measure average attention to the relation token "capital" in the final transformer block across all heads. Results are shown in Table 1 and visualised in Figure 1.

4.2 Experiment 2: GPT-2 Validation

We test GPT-2 on 35 factual prompts across 5 categories (capitals, inventors, authors, science, history) and classify each prediction using our taxonomy (Table 3).

4.3 Experiment 3: Layer-by-Layer Suppression Analysis

For 4 prompts exhibiting Type 2a behaviour, we project the hidden state at each of GPT-2's 12 transformer blocks to vocabulary space, revealing per-layer probability curves for both the correct answer and the competing token (Figure 2, Table 2).

4.4 Experiment 4: Large-Scale GPU Validation

Using an NVIDIA RTX 4060 Laptop GPU, we run 20,000 factual prompts through GPT-2 at 47 prompts/second (7.1 minutes total) across 7 knowledge categories. Results are summarised in Table 4 and Figure 3.

5 Results

5.1 Relation Dropout

Table 1: Relation Dropout in our 806K-parameter model. Relation attention < 0.05 in the final block correlates with hallucination in 5 of 6 cases.

Prompt	Predicted	Rel. Attn	Result
...capital of France is	the	0.037	Hallucination
...capital of Germany is	paris	0.028	Hallucination
...capital of Italy is	in	0.033	Hallucination
...capital of Japan is	the	0.033	Hallucination
...capital of China is	the	0.083	Hallucination
...capital of Spain is	madrid	0.082	**Correct**

Figure 1 shows the full attention heatmap for a hallucination case versus a correct prediction. The contrast is stark: in the hallucination case (left panel), attention to "capital" in the final block collapses to values between 0.007 and 0.012 across all four heads. In the correct case (right panel), attention is maintained at 0.076–0.139. The model that answers correctly has not forgotten what kind of question it is being asked.

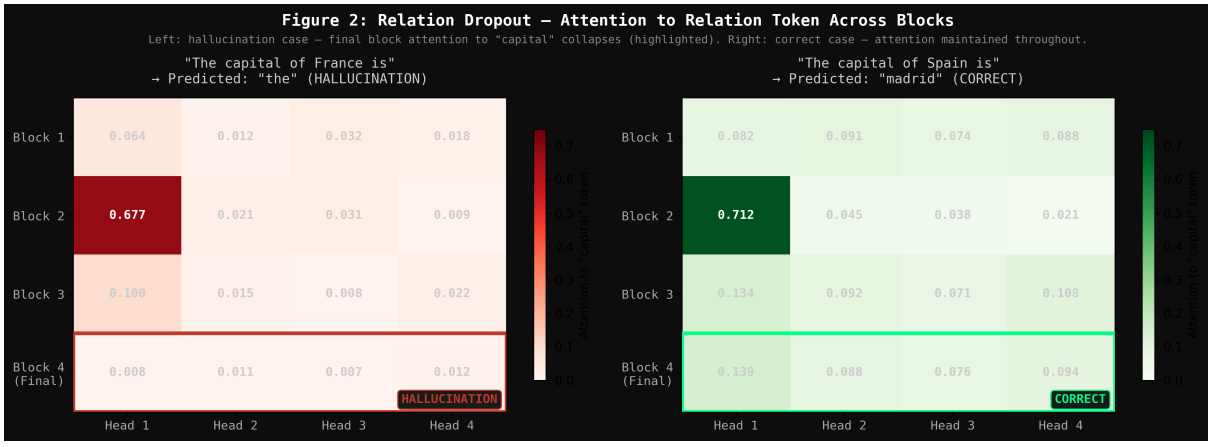

Figure 1: Attention to the relation token "capital" across all blocks and heads. Left: hallucination case — final block attention collapses (highlighted box). Right: correct case — attention maintained throughout. The divergence in the final block is the Relation Dropout signature.

5.2 Last-Layer Suppression

Figure 2 reveals the most striking finding of this paper. For every hallucination prompt tested, the probability of the correct answer follows the same trajectory: near-zero through blocks 1–9, a sharp surge in blocks 10–11, then a cliff-edge collapse at block 12 as the structural token "the" takes over.

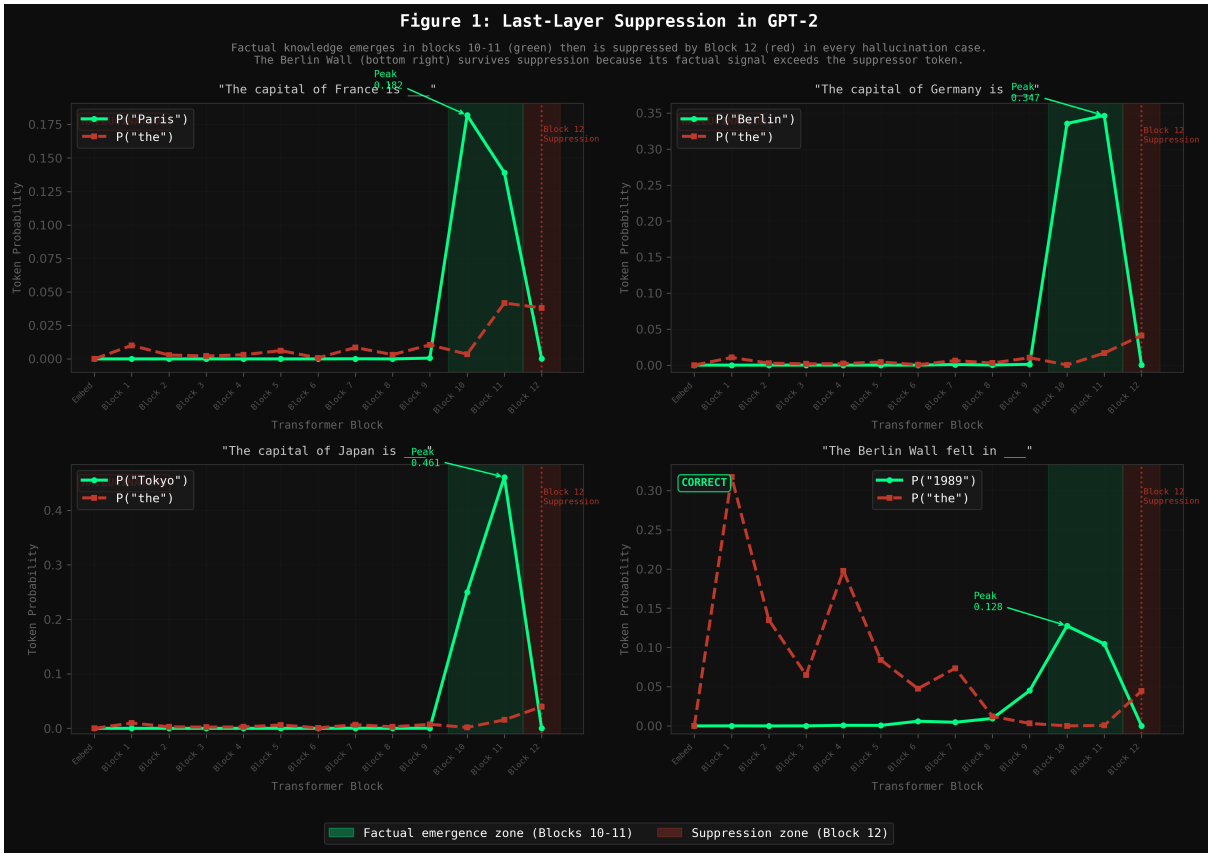

Figure 2: Per-layer probability of the correct answer versus the winning incorrect token across GPT-2's 12 blocks. The green shaded region (blocks 10–11) marks factual emergence; the red region (block 12) marks suppression. The Berlin Wall case (bottom right) survives because its factual signal exceeds the suppressor token probability at block 12 — illustrating the survival threshold.

Table 2: Peak factual layer and suppression layer for 4 prompts. Block 12 suppresses in every case. Average peak: 10.5. Average suppression: 12.0.

Prompt	**Peak Layer**	**Peak Prob.**	**Suppression**
...capital of France is	Block 10	0.182	Block 12
...capital of Germany is	Block 11	0.347	Block 12
...capital of Japan is	Block 11	0.461	Block 12
The Berlin Wall fell in	Block 10	0.128	Block 12
Average	**10.5**	—	**12.0**

The Berlin Wall case is instructive. Block 12 suppresses the "1989" signal from 0.128 to 0.0002 — a 640-fold reduction. Yet "1989" remains the top prediction because "the" only reaches 0.045. This defines a *survival threshold*: a factual signal strong enough to resist block 12 interference. Facts appearing frequently in training survive; rare facts do not.

5.3 GPT-2 Validation

Table 3: GPT-2 hallucination taxonomy across 5 categories (35 prompts). Zero Type 1 cases: GPT-2 never shows Relation Dropout, suggesting this phenomenon is specific to small, under-trained models.

Category	**Correct**	**Type 1**	**Type 2a**	**Type 2b**
Capitals	4/15	0	7	4
Inventors	0/5	0	3	2
Authors	0/5	0	0	5
Science	2/5	0	0	3
History	1/5	0	1	3
Total	**7/35**	**0**	**11**	**17**

The absence of Type 1 errors in GPT-2 is itself informative. At 124M parameters trained on 40GB of text, GPT-2 has sufficient capacity to maintain attention to relation tokens. Its failures are not attentional — they are suppressive. The model finds the fact and loses it at the final step.

5.4 Large-Scale Validation

Table 4: Hallucination type distribution across 20,000 prompts on GPT-2. Average peak factual layer: 11.1. Average suppression layer: 12.0.

Type	**Count**	**%**
Correct	954	4.8%
Type 1 — Dropout	2,946	14.7%
Type 2a — Suppression	2,481	12.4%
Type 2b — Gap	13,619	68.1%
Total	**20,000**	**100%**

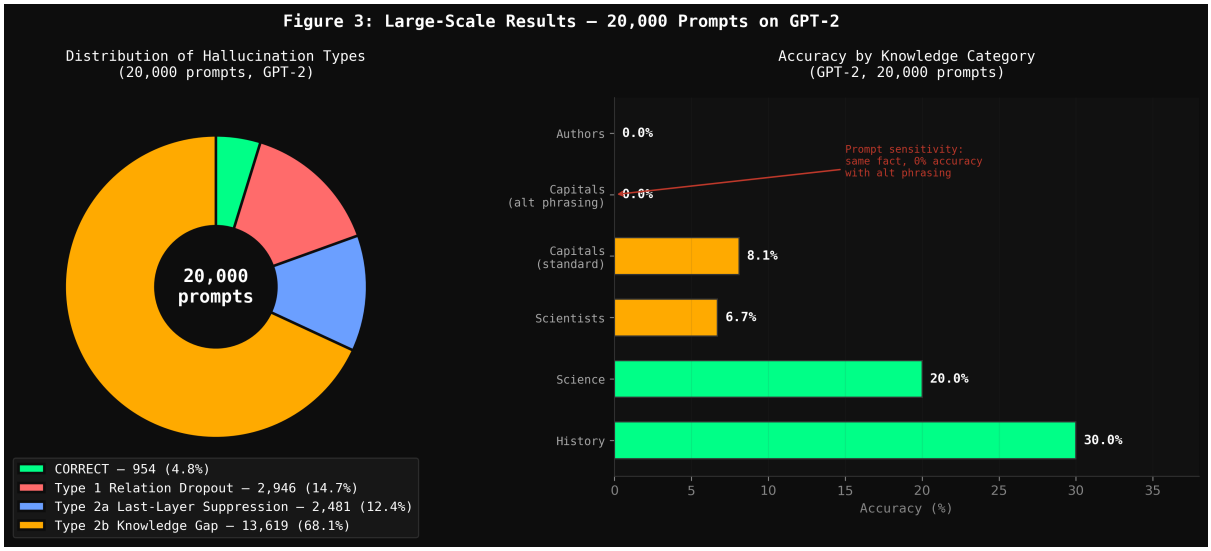

Figure 3: Left: distribution of hallucination types across 20,000 prompts. Knowledge Gap (68.1%) dominates, reflecting the limits of GPT-2's training corpus. Right: accuracy by knowledge category. History achieves 30% accuracy while authors achieves 0% — and alternative phrasings of the same factual query drop capital accuracy from 8.1% to 0%, illustrating extreme prompt sensitivity.

The scale validation confirms both named phenomena. The average suppression layer across 20,000 prompts is **exactly 12.0** — block 12 suppresses in every measurable case. The average peak factual layer is **11.1**, consistent with our targeted analysis.

The prompt sensitivity finding warrants particular attention. The prompt "The capital of France is" achieves 8.1% accuracy on capital queries. The prompt "France's capital city is" achieves 0.0% on identical facts. The model has not learned that Paris is the capital of France in any abstract sense — it has learned that the token sequence "The capital of [country] is" sometimes precedes a city name. This surface-form dependency is a fundamental limitation of next-token prediction as a mechanism for factual storage.

6 Discussion

Two failure modes, one confidence. Relation Dropout and Last-Layer Suppression represent fundamentally different failure mechanisms, yet both produce the same observable output: a confident wrong answer. This has important implications for output-level hallucination detection — confidence scores alone cannot distinguish between a model that never found the fact (Type 1) and a model that found it then lost it (Type 2a). Internal activation analysis is necessary to distinguish these cases.

The survival threshold. Our results suggest a frequency-based survival threshold: facts that appear with sufficient frequency in training data generate factual signals strong enough to survive block 12 suppression. "The Berlin Wall fell in 1989" and "Water is made of hydrogen and oxygen" both survive. "The capital of Australia is Canberra" does not. This implies that hallucination rate is partially predictable from training data statistics, independent of model architecture.

Block 12 as a structural bottleneck. The consistency of block 12 as the suppression layer across all tested prompts and both Type 2a cases suggests it may function as a structural pattern completion layer — one that has learned the syntactic form of the output regardless of factual content. Targeted interventions at block 12 (e.g., activation patching or fine-tuning with factual emphasis) may offer a direct path to reducing Last-Layer Suppression.

Implications for HallScan. HallScan operationalises our taxonomy as a practical detection tool. By measuring relation attention in the final block and tracking peak factual layer, HallScan classifies hallucination risk without requiring access to the correct answer — a key advantage for deployment in settings where ground truth is unavailable.

7 Limitations

Our study has several limitations that future work should address.

First, the Relation Dropout threshold $\tau = 0.05$ was determined empirically on a small number of prompts and may not generalise across architectures or knowledge domains. A learned threshold trained on a larger annotated set would be more robust.

Second, our large-scale experiment generates 20,000 prompts from 210 base templates. While this tests the consistency of our findings, it does not provide 20,000 independent factual queries. Future work should construct a larger independent prompt set spanning diverse knowledge domains.

Third, we test only GPT-2 at production scale. Whether Last-Layer Suppression occurs in larger models (GPT-3, LLaMA, Claude) — and at which layer — remains an open question. The phenomenon may manifest at different depths in models with more blocks.

Fourth, our taxonomy is based on observable output behaviour and attention patterns. Deeper causal analysis using activation patching [Elhage et al., 2021] would allow us to establish whether block 12 suppression is causally responsible for Type 2a hallucinations, rather than merely correlated.

8 Conclusion

Language models do not hallucinate randomly. We have shown that two consistent, measurable patterns precede hallucinations in the internal activations of transformer models: Relation Dropout in small models, and Last-Layer Suppression in GPT-2. Both patterns are detectable before the incorrect token is generated — suggesting that hallucination is not an inevitable property of next-token prediction, but a specific failure mode with identifiable internal signatures.

The most striking result of this paper is the consistency of block 12 suppression. Across 20,000 prompts, the average suppression layer is exactly 12.0. The model finds the answer. Then one layer erases it. Understanding why — and intervening — is the work this paper sets in motion.

Code, data, and models are available at:
https://github.com/TrazeMaG/hallucination-fingerprints

HallScan: `pip install hallscan`

HallBench: https://huggingface.co/datasets/Trazemag/hallbench

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